Research Vision

Magnetic Nanotechnology for Real-Time Imaging, Diagnostics, and Therapy

Overall Objective and Long-Term Goal

The overall objective of my research program is to develop magnetic nanotechnologies that enable real-time, non-invasive imaging and diagnostics to improve the detection, monitoring, and treatment of cancer and infectious diseases. My long-term goal is to establish an independent research program that integrates materials chemistry, biomedical imaging, and translational diagnostics to replace invasive, delayed, and infrastructure-heavy workflows with adaptive, magnetic nanotechnology–driven platforms.

This research is driven by the premise that **diagnostic capability must evolve in parallel with therapeutic innovation**. While cancer immunotherapies and antimicrobial treatments have advanced rapidly, diagnostic tools capable of providing actionable, real-time feedback remain limited. Magnetic nanotechnology offers a unique opportunity to close this gap by enabling molecularly specific, quantitative imaging and detection without ionizing radiation and with inherent scalability to point-of-care formats.

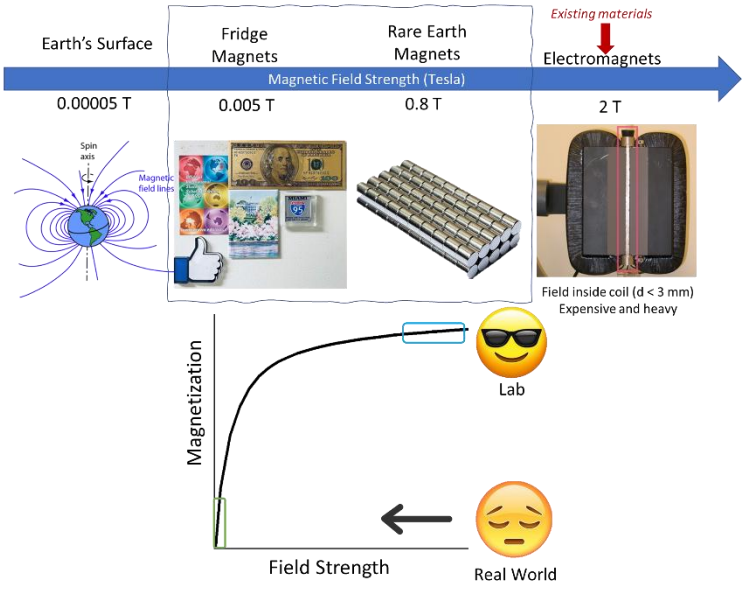
Significance

Cancer immunotherapy has transformed oncology, yet clinicians currently lack tools to non-invasively and quantitatively assess treatment response early in the therapeutic course. As a result, ineffective therapies are often continued for prolonged periods, exposing patients to unnecessary toxicity and delaying alternative interventions. Similarly, the global rise of antimicrobial resistance has outpaced diagnostic infrastructure, particularly in resource-limited settings where culture-based and PCR diagnostics are slow, expensive, or unavailable.

These gaps represent critical unmet clinical needs. Real-time, non-invasive diagnostics capable of informing treatment decisions would fundamentally change how therapies are deployed in both oncology and infectious disease. Magnetic nanotechnology is uniquely positioned to address these challenges due to its non-ionizing nature, compatibility with portable instrumentation, and tunable molecular specificity. The proposed research directly addresses these needs by creating magnetic nanomaterial platforms that translate fundamental materials innovation into clinically actionable diagnostic and imaging tools.

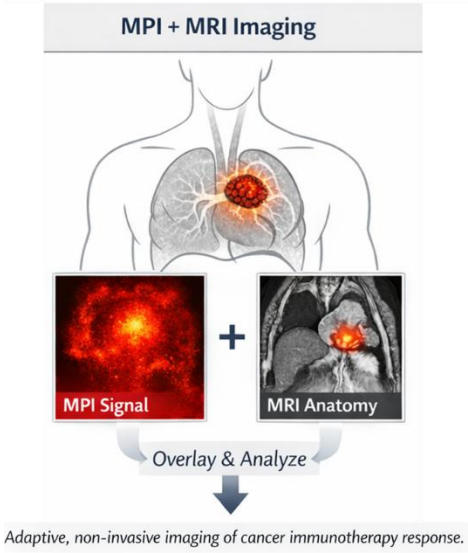
Innovation

1. **Ultra-low-field magnetic responsiveness**
   By engineering magnetic nanomaterials with enhanced susceptibility and tailored anisotropy, this work advances materials capable of responding to ultra-low magnetic fields. This eliminates dependence on expensive superconducting magnets and expands access to magnetic diagnostics and therapies.

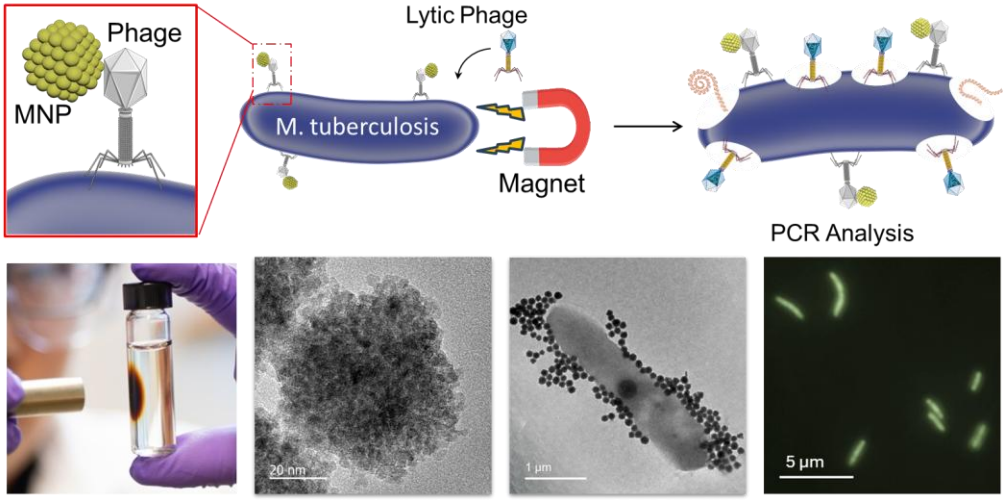


Challenges of nanotechnology beyond laboratory

2. **Activatable magnetic contrast mechanisms**
   Unlike conventional magnetic nanoparticles that provide static contrast, my approach introduces stimulus-responsive nanoprobes whose magnetic relaxation properties switch in direct response to disease-relevant biomarkers, including cytokines, reactive oxygen species, and tumor microenvironment pH. This enables imaging readouts that reflect biological function rather than passive accumulation.



3. **Integration of imaging, diagnostics, and translation**
   The research bridges magnetic particle imaging (MPI), MRI, and portable magnetic diagnostics within a unified materials framework. This integration enables seamless translation from preclinical imaging to point-of-care detection, a gap rarely addressed in a single research program.



Research Strategy and Approach

My research strategy is organized around **three interconnected platform technologies** that support both fundamental discovery and translational deployment.

Platform 1: Magnetic Nanomaterials with Enhanced Susceptibility

I will design and synthesize reusable magnetic nanoparticle architectures with optimized crystal phase, assembly, and surface chemistry to achieve enhanced magnetic susceptibility and tunable relaxation behavior. These materials will be characterized using magnetometry, MRI, and MPI to establish structure–property relationships that guide rational design. This platform provides the materials foundation for all subsequent imaging and diagnostic applications.

Platform 2: Activatable MPI/MRI Nanoprobes for Precision Oncology

Building on these materials, I will develop activatable nanoprobes whose magnetic signals switch in response to immunological biomarkers associated with effective tumor killing. These probes will leverage the zero-background sensitivity of MPI combined with anatomical MRI co-registration to enable quantitative, longitudinal monitoring of immunotherapy response. Validation will be performed in syngeneic tumor models to establish biological specificity, sensitivity, and temporal resolution.

Platform 3: Translational Magnetic Diagnostics for Global Health

To address infectious disease diagnostics, I will translate magnetic nanomaterials into portable, field-deployable platforms. Magneto-bacteriophage conjugates will be engineered to selectively capture and concentrate viable pathogens from clinical samples, enabling rapid, culture-free detection. Multiplexed phage libraries targeting WHO-priority antimicrobial-resistant organisms will be evaluated under field-simulated conditions to assess robustness and scalability.

Expected Outcomes and Impact

- New classes of activatable magnetic nanoprobes for functional imaging
- Materials design rules for ultra-low-field magnetic responsiveness
- Portable diagnostic platforms enabling rapid pathogen detection at the point of care

Collectively, these outcomes will:

- Enable adaptive, real-time treatment decisions in cancer immunotherapy
- Reduce reliance on invasive biopsies and delayed clinical endpoints
- Improve access to diagnostics in resource-limited settings
- Support antimicrobial stewardship by enabling earlier identification of resistant infections

Beyond clinical impact, this work will contribute broadly to the fields of nanomaterials, imaging science, and translational diagnostics, while training students at the interface of chemistry, engineering, and medicine.

Long-Term Vision

My long-term goal is to establish an independent research program that reshapes how cancer and infectious diseases are diagnosed and monitored by replacing invasive, delayed workflows with real-time magnetic nanotechnology–driven platforms. By uniting fundamental materials innovation with imaging science and translational deployment, this work will advance both scientific understanding and public health impact.